Houndou Bongo Koyna, Tombouctou

Here the danger is the calendar: boats stop landing grain, medicine, and fuel in January, so everything the village needs until the rains must be landed at Koriomé by mid-December.

REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Tombouctou Region (region level)	170 forecast, 61 of them in March; 200 already past (Jun–Sep)	6-month forecast, Oct 2026 – Mar 2027	The river closing to boats under a blockade; malnutrition; meningitis Jan–May; heat from February

27/27 recommendations checked. 8 confirmed, 17 corrected, 0 unverified, 2 contraindicated. That covers the 9 actions (1 confirmed, 8 corrected), the 8 "Do not do this" cards (4 confirmed, 2 corrected, 2 contraindicated, one of which began as a draft resident step), and the 10 For you steps (3 confirmed, 7 corrected). Exposure figure: Climate Impact Lab El Niño Heat Mortality Tracker data, version of 26 October 2026, pre-peer-review. It is published for Tombouctou Region, not for the village, and it counts heat only. Nothing published covers Houndou Bongo Koyna itself, so the actions are built for Timbuktu town, its river ports, and the region, and apply to the village by proximity. The funding target moved from the panel's XOF 375 million (US$638,600) to XOF 316 million (US$538,600)

EXECUTIVE SUMMARY — AT A GLANCE



16.66°N 2.96°W, Mali · Natural Earth

- **The risk.** Boats reach Timbuktu's ports only from July to January; after that, supply comes by road through a blockade that raised food prices almost tenfold in 2023.
- **For you.** Before mid-December, buy two to three months of grain while the boats still land it, and store it in sealed bags.
- **The money.** XOF 105–651 million (US$178,100–1.1 million); malnutrition treatment landed by 15 December buys the most lives.
- **Do not.** Do not put soldiers on civilian passenger boats: witnesses counted at least 120 dead when the *Tombouctou* was attacked in 2023.

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* Suggested formulation: check with a health worker before sharing.

after verification (−15.7%): the solar-borehole line has no named recipient and moved to the ceiling.

FOR YOU, AS A RESIDENT — 10 STEPS

For a family in Houndou Bongo Koyna, this winter's danger is less the heat than the calendar. The boats that bring grain, medicine, and fuel stop in January, and meningitis, the hungry months, and the heat all arrive after they stop. The ranked actions below are written for the region and its partners. These are the ones you can take yourself.

Have every child aged 6 months to 5 years measured with a MUAC arm band at the health centre (CSCom) this month

PARTIALLY CONFIRMED · FROM ACTION 2

BY WHEN	COST TO YOU	PROTECTS
November 2026	screening is free; treatment is usually free	young children

A child who is already wasted is the one most likely to die of fever, diarrhoea, or a hot day. Treatment at home with therapeutic food is given free where UNICEF or an NGO supplies it, and ICRC-supported health centres near Timbuktu treat children under five free. No national decree making it free was found, so ask. The arm band is not valid for babies under 6 months.*

Fever with a stiff neck, a purple rash that does not fade, or a baby with a bulging soft spot: go to the CSCom the same day

CONFIRMED · FROM ACTION 4

BY WHEN	COST TO YOU	PROTECTS
January–May 2027	transport	children and young adults

These are the signs of meningitis, and its season here runs from January to May. Mali-Météo put Tombouctou on meningitis climate alert in March 2025, and in 2018 only 34% of the region's children had the full basic vaccine series. Being vaccinated does not rule it out: MenAfriVac protects against group A only.*

Buy two to three months of grain before mid-December, while the boats still land it, and store it in sealed bags

* Suggested formulation: check with a health worker before sharing.

PARTIALLY CONFIRMED FROM ACTION 1

BY WHEN	COST TO YOU	PROTECTS
15 December 2026	the grain (no current Timbuktu price found), plus about XOF 1,200–1,800 (US$2–3) per triple-layer bag	food from February to June

Boats dock at Koriomé or Kabara only from July to January, during the flood. After that, food comes by road through the blockade, which raised prices almost tenfold in 2023. Hermetic triple-layer (PICS-type) bags cost about XOF 1,200–1,800 (US$2–3) each in Purdue's 2012–16 studies; whether they are sold in Timbuktu was not confirmed, so sealed jars or drums will do.

The day you hear the Timbuktu power plant has stopped, fill every container, and keep the water in covered jars

PARTIALLY CONFIRMED FROM ACTION 3

BY WHEN	COST TO YOU	PROTECTS
the same day	none	drinking water

The plant stopped on 19 June 2026, and the town lost power and piped water from 23 June for more than a week, in heat above 40°C. The village's own water source is not on record. Whatever it is, demand from town will crowd your well or borehole too.

Agree now with one neighbour who will fetch water for any old or sick person in your compound

PARTIALLY CONFIRMED ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
before 1 March 2027	none	people who cannot carry water

In June 2026, women queued for hours and walked far from home with their buckets. People who cannot queue go without, and in the heat that is how they die unseen.

No night crossings, and no overloaded pinasse or pirogue, until the water drops

* Suggested formulation: check with a health worker before sharing.

CONFIRMED FROM ACTION 5

BY WHEN	COST TO YOU	PROTECTS
now until about February 2027	none	everyone who travels by river

On 8 January 2026, at about 10 pm, a pinasse overloaded with sacks of rice sank off Diré with about 50 people aboard. Thirty-three bodies were recovered and 11 people survived. The governor's office urged boat operators to respect boarding rules, especially at night.

If the river rises, raise grain and seed off the floor, put earth bags on the river-facing wall, and ask the Croix-Rouge volunteers whether a flood alert is out

PARTIALLY CONFIRMED FROM ACTION 6

BY WHEN	COST TO YOU	PROTECTS
November 2026 – January 2027; roof drains before the rains	your own labour	grain, seed, and the house

In 2024, Tombouctou had 196 of Mali's 591 flood incidents, more than any other region, and high water reached Timbuktu and Diré from mid-November. Ask the volunteers whether an alert is out for the Diré–Kabara stretch and where sandbags are handed out. Clear roof drains before the rains as well: in 2003 it was August rain, not the river, that brought down banco houses in Timbuktu.

When the heat returns, use shade, water on the skin, and drink; no fan for older people above about 35°C

PARTIALLY CONFIRMED FROM ACTION 7

BY WHEN	COST TO YOU	PROTECTS
February–March 2027	none; keep ORS sachets (price here not found)	older people and anyone sick

The tracker puts 61 of the region's 170 forecast heat deaths in March. A 2024 review advises against fans above 35°C, especially for people over 65, because in dry heat a fan can warm the body. Wet the skin first and let the air cool it. There is often no grid power anyway.*

* Suggested formulation: check with a health worker before sharing.

Ramadan, about 8 February – 9 March 2027: the sick are exempt from fasting, and older people who cannot fast may pay fidya instead

CONFIRMED FROM ACTION 7

BY WHEN	COST TO YOU	PROTECTS
about 8 February 2027, depending on the moon	fidya, for those who pay it	older and sick people

The fast ends as the heat returns. Qur'an 2:184 exempts the sick, and people too old to fast pay fidya. In April 2024, during Ramadan and power cuts, Gabriel Touré hospital in Bamako recorded 102 deaths in four days, about half of them people over 60; the cause was not established.*

If you keep animals, sell the weak ones by December, before the pasture runs out

PARTIALLY CONFIRMED FROM ACTION 8

BY WHEN	COST TO YOU	PROTECTS
31 December 2026	none; the sale buys grain	the herd's value and the family's food

FEWS NET reported very severe pasture deficits in Tombouctou's main pastoral areas in September 2026, with water points 10–50% below normal. The droughts of 1973 and 1984–85 killed up to two-thirds of herds. No 2026 livestock price was found, so the timing rests on the pasture, not on the market.

What to ask of the people in charge

- Ask the Direction régionale de la santé (DRS) Tombouctou: "Will therapeutic food, ceftriaxone, and malaria tests be landed at our CSCom before the boats stop?" (actions 1, 2, and 4).
- Ask the governor and the boat owners: "Will night crossings stop, and will life jackets be carried, until the water drops?" (action 5).
- Ask the Croix-Rouge malienne and the national hydrology office (DNH): "Will a Diré or Kabara threshold and a Tombouctou stock be added to the flood protocol?" (action 6).

* Suggested formulation: check with a health worker before sharing.

WHAT THIS PLACE SHOULD DO RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet.

1. Land the dry-season stocks at Koriomé before the boats stop: cereal, therapeutic food, essential drugs, and about 6,000 litres of hospital diesel, by 15 December

PARTIALLY CONFIRMED | EMERGENCY MANAGER · FOOD · SKEPTIC · PHILANTHROPIST · HISTORIAN | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

OWNER	BY WHEN	COST	PREVENTS	PAYER / TRIGGER
Governor of Tombouctou, with the CSA, WFP, and the DRS	landed by 15 December 2026	freight not quoted; get quotes first	a road-only supply line through the blockade from February to June	the State (CSA, PNR 2026) and WFP; trigger: the date

Boats dock at Koriomé or Kabara only from July to January, during the flood (IMPACT, December 2023). The route is road from Bamako to Mopti, then about two weeks by pinasse to Koriomé. The Kabara channel is badly silted and passable only a few weeks a year, so Koriomé is the main port. After January, supply comes by road through the blockade in place since August 2023, which raised food prices almost tenfold that year (Save the Children). In Timbuktu, 49% of traders closed at some point, and prices rose 70–100% in three months. The drug shipment should carry malaria rapid tests, ACTs, and injectable artesunate. Test before treating, because many fevers here are not malaria and a stiff neck must not be treated as one.

* Suggested formulation: check with a health worker before sharing.

Corrected in verification: no evidence was found that WFP or CSA pre-positioned by river in 2024–26, so this is a proposal. No freight rate was published, only tourist pinasse charters at XOF 300,000–500,000 (US$511–852) a trip. The panel's XOF 400,000 (US$681) per tonne, and XOF 30 million (US$51,000) per 1,000 people for six months of cereal, are unsourced estimates. No 2026 flood bulletin for Diré or Tombouctou was found. The flood normally peaks here in December–January, and a low flood shortens the window.

2. Get therapeutic food to every CSCom and screen every child aged 6–59 months before January

PARTIALLY CONFIRMED | PUBLIC HEALTH · PHILANTHROPIST | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

OWNER	BY WHEN	COST	PREVENTS	PAYER / TRIGGER
DRS Tombouctou and the CSComs, with UNICEF	stock landed by 15 December 2026; screening before January	XOF 176.1 million (US$300,000) per 1,000 children	deaths of wasted children from heat, fever, and diarrhoea	UNICEF and donors; trigger: the date

Community treatment of severe acute malnutrition brought mortality to 4.1% across 21 programmes (23,511 cases), against 20–30% in older inpatient care (Collins et al., *Food and Nutrition Bulletin* 2006 supplement). In Mali, UNICEF reports 0.3% mortality among treated children (Q1 2026). An IPC projection for 2024 put about 65,000 acutely malnourished children in Tombouctou region, about 15,000 of them severe. Nationally, 1.12 million children are projected acutely malnourished from November 2025 to October 2026, 227,000 severely, with Tombouctou among the worst. UNICEF's 2026 appeal for Mali is 69% unfunded, a XOF 43.8 billion (US$74.6 million) gap, and Mali ran out of therapeutic food from late July 2024.

* Suggested formulation: check with a health worker before sharing.

Corrected in verification: the "12% in 2024" and "11.1% in 2025" prevalence figures had no source and were dropped. AFD's northern health facility is completed, and France suspended aid in 2022, so it is not a channel. MSF has no confirmed 2026 presence in Timbuktu town. The cost is a conservative delivered planning figure of XOF 176,100 (US$300) a child, not a Timbuktu figure: IRC's general range is XOF 58,700–293,600 (US$100–500) a child excluding the food, IRC in Nara, Mali, costs XOF 42,300–47,600 (US$72–81) excluding the food, and a carton (about one child's course) costs XOF 27,420 (US$46.70).

3. Keep the hospital and the water running through a fuel cut: land diesel at the Hôpital de Tombouctou before mid-December, survey the village's water, and visit every vulnerable compound within 24 hours of a shutdown

PARTIALLY CONFIRMED | PUBLIC HEALTH · FRONTLINE · EMERGENCY MANAGER · PHILANTHROPIST | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

OWNER	BY WHEN	COST	PREVENTS	PAYER / TRIGGER
DRS Tombouctou, with the Croix-Rouge malienne volunteers	**diesel landed by 15 December 2026**	**XOF 16.5 million (US$28,100) for about 6,000 L**	**wards going dark in the heat and meningitis season**	**donor grant to the DRS; visits on "the power plant stops"**

The Timbuktu power plant stopped on 19 June 2026 (EDM, via AFP). The town was without power and piped water from 23 June for more than a week, then had one to two hours a day per neighbourhood. Black-market petrol sold at XOF 2,500–3,000 (US$4.26–5.11) a litre, against the official XOF 875 (US$1.49). The hospital kept going on solar and generators. A UNICEF and BMZ solar system already covers maternity, neonatology, and paediatrics, so the diesel protects the other wards. The cost is 1,500 L a month for four months at XOF 2,750 (US$4.68) a litre.

* Suggested formulation: check with a health worker before sharing.

Corrected in verification: the volume is assumed; ask the DRS for the hospital's real consumption. At black-market prices, a 100–200 kVA generator running 12 hours a day for 30 days costs XOF 13.5–27 million (US$23,000–46,000). No water data exists for the village, which is about 12 km from the town network, so its source must be surveyed first. Solar boreholes have no published unit cost and no named NGO, so they are in the ceiling only.

4. Meningitis: put ceftriaxone, lumbar-puncture kits, and rapid tests in the CSComs by December, and report cases weekly

PARTIALLY CONFIRMED | PUBLIC HEALTH · SKEPTIC | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

OWNER	BY WHEN	COST	PREVENTS	PAYER / TRIGGER
DRS Tombouctou	stocked by 15 December 2026 (on action 1's shipment); weekly reports from January	inside the drug line: XOF 17.6 million (US$30,000), an assumption	deaths in the weeks before a reactive vaccine arrives	donor grant to the DRS; alert at 5 cases per 100,000 a week, or 2 a week under 30,000 people

Reactive vaccination starts a median of 36 days after a district crosses the epidemic threshold, so early treatment is what saves lives in the first weeks. Tombouctou lies on the dry northern edge of the meningitis belt, at about 200 mm of rain a year, and is not listed as high-risk. But Mali-Météo issued a meningitis climate alert for it in March 2025, and only 34% of its children had the full basic vaccine series in 2018, against 45% nationally. Ask the immunisation programme (PEV) whether the Men5CV vaccine has reached Tombouctou.

* Suggested formulation: check with a health worker before sharing.

Corrected in verification: no Mali Men5CV campaign or routine introduction was found for 2024–26, so the briefing does not claim one either way. The ceftriaxone protocol is cited from WHO epidemic guidance that could not be reopened in this run.

5. Make the river crossings survivable: no soldiers on civilian passenger boats, and no night travel, load limits, and life jackets during high water

CONFIRMED | EMERGENCY MANAGER · HISTORIAN | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

OWNER	BY WHEN	COST	PREVENTS	PAYER / TRIGGER
COMANAV and the Ministry of Transport; the governor with boat owners	now, until the water drops (about February 2027)	XOF 0	a repeat of the 2023 attack and the 2026 Diré sinking	no money needed; a decision

On 7 September 2023, the COMANAV boat *Tombouctou* was attacked near Banikane/Zarho with a few soldiers aboard as escort, who returned fire. Human Rights Watch's witnesses put the dead on the boat at 120 or more; one survivor counted 154 bodies, and UNICEF counted 24 children. On 8 January 2026, at about 10 pm, a pinasse overloaded with rice sank off Diré: about 50 aboard, 33 bodies recovered, 11 survivors. The governor's office urged respect for boarding rules, especially at night.

Corrected in verification: the government's toll of 49 civilians and 15 soldiers combined the boat with a separate attack on the Bamba camp, and the claim that the boat was rated for 300 and carried 500 or more was dropped as unsourced. No order banning night travel or requiring life jackets exists, so this is an agreement to seek, not a rule to enforce.

6. Watch the river both ways, and get a Timbuktu trigger and stock into the Red Cross flood protocol

PARTIALLY CONFIRMED | ECONOMIST · EMERGENCY MANAGER · HISTORIAN · SKEPTIC | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

OWNER	BY WHEN	COST	PREVENTS	PAYER / TRIGGER
Croix-Rouge malienne and DNH; Civil Protection and the mayors for the watch	watch November 2026 – January 2027; protocol at its next annual review	about XOF 3 million (US$5,110) for sandbags and 20 volunteers, an estimate	a high flood with no stock nearby, and a low flood nobody sees	IFRC DREF via the Croix-Rouge malienne; a forecast 5-year flood level 4–8 days ahead

* Suggested formulation: check with a health worker before sharing.

Mali's IFRC flood early-action protocol (MDRML021, approved 14 March 2025, valid to 31 March 2030) holds CHF 493,567 (about XOF 349 million, US$594,000), with CHF 187,236 (about XOF 132 million, US$225,000) for early actions for 4,000 households, or 24,000 people. It pays in kind: sandbags, shelter, water and health kits, nets, and evacuation. It names only two trigger stations, Bamako (370 cm) and Kayes (859 cm), and pre-positions stock in Kayes, Sikasso, Mopti, and Gao, not Tombouctou. It was not activated in 2025; the last activation was in Mopti in September 2022. Run a riverbank watch at Koriomé, Kabara, and Day, and a low-flood watch too, because a low flood cuts recession rice, bourgou pasture, and fish (report it to the CSA).

Corrected in verification: Tombouctou is named only in the protocol's list of high-risk areas. No 2026 flood forecast is published, so the flood cannot be called high or low. In 2024 the river at Mopti reached 5.90 m on 20 August, the highest since 1967; Tombouctou region had 36,317 people affected by 9 September and 196 of Mali's 591 flood incidents; Mali counted 86–95 dead. A government riverine-flood framework, XOF 881 million (US$1.5 million) for 16,000 people, is listed as active, but no trigger or area was found. The sandbag cost is an unsourced estimate.

7. A heat plan that works without power, in Songhai and Tamasheq, before Ramadan: radio messages, briefed imams, and volunteer compound lists

PARTIALLY CONFIRMED | FRONTLINE · PUBLIC HEALTH · SKEPTIC | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

OWNER	BY WHEN	COST	PREVENTS	PAYER / TRIGGER
Croix-Rouge malienne, with Studio Tamani, Radio Bouctou, and the imams	lists by 31 January 2027; imams briefed before about 8 February	XOF 6.0 million (US$10,200) for 50 volunteers over 4 months	older and sick people dying unseen in hot banco rooms	donor grant to the Croix-Rouge; Unfunded

Studio Tamani reaches 82% of over-15s in Timbuktu (Fondation Hirondelle, 2025) and broadcasts in Bambara, Peul, Tamasheq, Songhai, and French. Two-minute messages in Koyra Chiini and Tamasheq: shade, water on the skin, drink, no fans above about 35°C for people over 65. Imams can remind people before Ramadan (about 8 February – 9 March 2027) that the sick are exempt and the frail elderly may pay fidya. Volunteers list who lives alone, who sleeps in a hot banco room, who hosts displaced guests, and who has no radio. In the April 2024 Sahel heat, Kayes reached 48.5°C, and Gabriel Touré hospital in Bamako recorded 102 deaths in four days, about half over 60.*

Corrected in verification: Studio Tamani does not broadcast in Hassaniya, so Hassaniya messages need another station. Radio Bouctou's current on-air status is unverified. No statement by the Haut Conseil Islamique on fasting in heat was found, so briefing imams is a proposal. The cause of the Bamako deaths was not established. The XOF 30,000 (US$51) monthly stipend is assumed. Skeptic's warning:

* Suggested formulation: check with a health worker before sharing.

money spent on heat before February is probably spent on the wrong hazard, so this plan is cheap and dated for February on purpose.

8. Herders: claim Tombouctou's share of the national feed plan, and sell weak animals by December

PARTIALLY CONFIRMED | ECONOMIST · HISTORIAN · FOOD | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

OWNER	BY WHEN	COST	PREVENTS	PAYER / TRIGGER
CSA, with the governor	allocation asked for in November; released before the March lean season	administrative	herds dying before the pastoral lean season	Government of Mali (PNR 2026); already validated

FEWS NET (September 2026) reports very severe pasture deficits in Tombouctou's main pastoral areas and water points 10–50% below median, with insecure Tombouctou projected to improve from Phase 3 to Phase 2 from October. The national response plan (PNR 2026, validated in April, launched 11 August) covers 1,560,189 people with about 20,000 t of free cereal and livestock feed, but publishes no regional split. Ask the CSA for the Tombouctou share.

* Suggested formulation: check with a health worker before sharing.

Corrected in verification: the PNR budget is reported both as XOF 127 billion (US$216 million) and as XOF 12.7 billion (US$21.6 million), and neither is confirmed. No 2026 livestock-to-grain terms of trade were found; the last published figure was –16% against the five-year average in May 2024.

9. Give traders stock credit in November, and cash from March only where goods are in the market; in-kind food where they are not

PARTIALLY CONFIRMED | FOOD · SKEPTIC · PHILANTHROPIST | ADAPTATION INVENTORY: NO PRECEDENT CONFIRMED

OWNER	BY WHEN	COST	PREVENTS	PAYER / TRIGGER
WFP, with the CSA	credit in November 2026; lean-season support March–June 2027	XOF 100.0 million (US$170,300) for 500 households over 4 months, plus about XOF 2 million (US$3,400) of credit per trader	empty shelves and hunger in the road-only months	WFP and donors; trigger: goods in the market

WFP resumed life-saving aid for 40,000 people in Mopti and Timbuktu on 30 April 2026 and needs XOF 18.0 billion (US$30.6 million) for May–October. National food aid was 14.9% funded on 25 August (FEWS NET). Under the 2023 siege prices rose almost tenfold, and when goods cannot get in, cash only pushes prices up. The support is 500 households × 4 months × XOF 50,000 (US$85), in kind or mixed where markets are empty.

Corrected in verification: no public monthly Timbuktu price series exists (GIEWS, 27 August 2026, says only that prices are "significantly higher" than elsewhere), so index transfers to WFP or OMA data obtained directly. The XOF 50,000 transfer and the trader credit are unverified estimates. The window moved to March–June to match the pastoral lean season.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Governor of Tombouctou	DRS Tombouctou and CSComs	CSA	WFP	UNICEF	COMANAV and Ministry of Transport	Boat owners	Croix-Rouge malienne	DNH	Civil Protection and mayors	Radio and imams
1. Land stocks at Koriomé by 15 Dec	**A**	**R**	**R**	**R**	**C**	**C**	**R**		**C**	**I**	
2. Therapeutic food and screening	**I**	**A/R**		**C**	**R**			**C**			
3. Hospital diesel, water, 24-hour visits	**C**	**A/R**			**C**			**R**		**I**	
4. Meningitis readiness	**I**	**A/R**			**C**			**I**			**I**
5. River-crossing rules	**R**					**A**	**R**			**C**	**I**
6. River watch and flood-protocol trigger	**C**		**I**					**A**	**R**	**R**	
7. Heat plan in Songhai and Tamasheq	**I**	**C**						**A/R**			**R**
8. Herders' feed share	**R**		**A**	**I**							
9. Trader credit and conditional cash	**C**		**C**	**A/R**							

What the chart shows: the DRS is accountable for three of the nine actions and carries the life-saving core, and the governor is accountable for the one that decides all the others, the December river shipment. The Croix-Rouge malienne carries two. Actions 5, 6, and 8 cost almost nothing; they need decisions from COMANAV, the Ministry of Transport, the Red Cross protocol review, and the CSA, not money.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs. It covers Timbuktu town, its ports, and the cercle; nothing is costed for the village alone.

FLOOR

XOF 104.6 million (US$178,100)

TARGET

XOF 316.2 million (US$538,600)

CEILING

XOF 651.1 million (US$1,108,900)

LARGEST UNCERTAINTY

whether freight and diesel can be bought and landed before mid-December

Floor = therapeutic treatment for 500 children (XOF 88.1 million, US$150,000) + hospital diesel (XOF 16.5 million, US$28,100) = XOF 104.6 million (US$178,100). Target = UNICEF and the CSComs for 1,000 children (XOF 176.1 million) + the hospital's diesel and drugs (XOF 34.1

* Suggested formulation: check with a health worker before sharing.

million) + WFP lean-season support (XOF 100.0 million) + Red Cross volunteers (XOF 6.0 million) = XOF 316.2 million (US$538,600). Ceiling = target + a further 1,000 children (XOF 176.1 million) + a second WFP tranche (XOF 100.0 million) + solar on five boreholes (XOF 58.7 million) = XOF 651.1 million (US$1,108,900). Beyond that, river freight before January runs out. The target moved −15.7% in verification, from XOF 374.9 million (US$638,600), when the solar line moved to the ceiling. Mali's 10% levy on funding to associations and foundations (decree 2026-0152, 13 March 2026) could add up to 10% to the Red Cross and NGO lines; whether it applies to UN agencies or the Red Cross is unconfirmed.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
UNICEF Mali, earmarked for the DRS Tombouctou CSComs	Action 2: treatment for 1,000 children, freight ring-fenced and committed by 15 December (floor: 500 children)	XOF 176.1 million (US$300,000)	direct grant (earmarked)	XOF 176,100 (US$300) per child delivered, a conservative planning figure; community treatment 4.1% mortality (Collins 2006)	Partially confirmed
Hôpital de Tombouctou, via the DRS	Actions 3 and 4: about 6,000 L of diesel landed by mid-December (floor), plus drugs	XOF 34.1 million (US$58,100): diesel XOF 16.5 million, drugs XOF 17.6 million	direct grant	Volumes assumed; diesel at the black-market midpoint of XOF 2,750 (US$4.68) a litre	Partially confirmed
WFP Mali, Timbuktu operation	Action 9: 500 households × 4 months, March–June, in kind or mixed	XOF 100.0 million (US$170,300)	top-up to an existing operation	WFP cut its caseload from six months to three, so this adds coverage; transfer value unverified	Partially confirmed
Croix-Rouge malienne, Tombouctou committee	Actions 3 and 7: 50 volunteers × 4 months, heat and meningitis outreach	XOF 6.0 million (US$10,200)	direct grant (+10% levy possible)	No cost-per-death study; cheap coverage; stipend assumed	Partially confirmed
UNICEF Mali, earmarked as above	Action 2: a further 1,000 children (ceiling only)	XOF 176.1 million (US$300,000)	direct grant (earmarked)	Released only if admissions and freight allow	Partially confirmed

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
WFP Mali, Timbuktu operation	Action 9: a second tranche (ceiling only)	XOF 100.0 million (US$170,300)	top-up to an existing operation	As above	Partially confirmed
An NGO to be named	Action 3: solar on 5 boreholes (ceiling only)	XOF 58.7 million (US$100,000)	direct grant, only once a named NGO quotes	No published unit cost; no named recipient	Unverified

Already in place, and thin: Mali's 2026 humanitarian plan was 23% funded on 28 September, XOF 79.6 billion (US$135.6 million) of XOF 339.3 billion (US$577.9 million), and more than 200,000 people went unassisted. CERF gave Mali about XOF 5.9 billion (US$10 million) in October 2026, from a XOF 93.9 billion (US$160 million) round; its regions are not known. Mali has no drought anticipatory framework (OCHA's June 2026 list omits it), and only the Humanitarian Coordinator in Bamako can request a CERF proactive allocation. No African Risk Capacity policy for Mali was found for 2025–27, and WFP's shared Sahel layer, XOF 17.6 billion (US$30 million) for 2024/25, was not found renewed. It reads June–September rain, so it cannot see a low river flood in December–January. The World Bank's Jigisemejiri safety net closed on 30 June 2023, so it is not a channel. UNHAS flies again but carried only 11 t of light cargo across seven destinations in March–April. The ICRC is still in Timbuktu.

A funder's view: pay for the freight explicitly, and commit before mid-December. Here the binding cost is fuel and access, not therapeutic food. Money committed after January arrives after the deaths.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here.

Telling residents to check whether they are "on the flood-protocol list"

CONTRAINDICATED

Found in verification; it began as a draft resident step. No standing household list exists. The Red Cross protocol chooses communities only after a trigger fires, gives help in kind, not cash, and has no Timbuktu gauge as a trigger. Asking about "the list" builds a false expectation of enrolment or cash.

Instead: if the river rises, ask the Croix-Rouge volunteers whether a flood alert is out for the Diré–Kabara stretch and where sandbags are handed out (action 6).

Running a local azithromycin mass distribution for children

CONTRAINDICATED

The AVENIR trial in Niger (*NEJM* 2024) cut child deaths by 14% (95% CI 7–22) among children aged 1–59 months. But the trial in Mali and Burkina Faso (Chandramohan et al., *NEJM* 2019, 19,578 children) found no added benefit on top of seasonal malaria chemoprevention, with a rate ratio of about 1.1 (0.88–1.3), and rising macrolide resistance. The LAKANA trial in Mali found no benefit in infants. WHO's 2020 guideline limits it to infants aged 1–11 months, with resistance monitoring. It needs twice-yearly census-based rounds that cannot run under the blockade before the peak.

* Suggested formulation: check with a health worker before sharing.

Instead: treat malnutrition (action 2) and stock meningitis treatment (action 4); leave azithromycin to a national policy decision.

Relying on cash alone where goods cannot get in

DO NOT CONFIRMED

Under the 2023 siege food prices rose almost tenfold. WFP's own 2012 preconditions for cash in the north (food in the markets, traders, open routes, banks, security) were not met then, and the river closes again in January. When supply is blocked, cash pushes prices up.

Instead: land the stocks first (action 1), then cash only where goods are in the market (action 9).

Putting soldiers on civilian passenger boats

DO NOT CONFIRMED

On 7 September 2023 the COMANAV boat *Tombouctou*, with a few soldiers aboard as escort, was attacked near Banikane/Zarho. Witnesses put the dead at 120 or more.

Instead: keep civilian boats civilian, and agree no night travel and load limits during high water (action 5).

Planning only for a repeat of the 2024 flood

DO NOT CONFIRMED

No 2026 flood forecast is published. El Niño tends to weaken the Sahel rains, but PRESASS 2026 forecast normal-to-above rain for the inner Delta. The flood here depends on rain over Guinea, and it could come high or low. A low flood cuts recession rice, bourgou pasture, and fish, and shortens the boat window.

* Suggested formulation: check with a health worker before sharing.

Instead: watch the Diré and Kabara gauges in November–December and set triggers both ways (action 6).

Waiting for a meningitis epidemic before preparing

DO NOT CONFIRMED

Reactive vaccination starts a median of 36 days after the epidemic threshold is crossed. After January, nothing reaches Timbuktu by river.

Instead: ceftriaxone and rapid tests in the CSComs by December, and weekly case reporting (action 4).

Telling people to travel or evacuate by convoy or boat

DO NOT PARTIALLY CONFIRMED

On the river and the road, movement is the hazard, not the way out: the 2023 boat attack and the January 2026 Diré sinking show it. No Timbuktu-specific failed evacuation is documented, so the evidence is indirect.

Instead: bring the stocks to the people before the peak (action 1).

Advising fans for older people in dry heat above about 35°C*

DO NOT PARTIALLY CONFIRMED

A 2024 review (Meade, Notley, Kirby, and Kenny, *Lancet Planetary Health*) advises against fans above 35°C, especially for people over 65. Modelling (Morris et al., 2021) puts the limit at about 39°C for young adults, 38°C for older adults, and 37°C for older adults on anticholinergic drugs, depending on humidity. A 2024 *JAMA* trial found fans did not lower older adults' peak core temperature. There is often no grid power anyway.

Instead: shade, water on the skin, and drinking water (action 7).

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1913–14	Sahel famine, remembered by Songhai speakers as the "great famine"	About 250,000 to 1 million dead across the Sahel	Nothing: the colonial state gave no relief
1973, 1984–85	Droughts; in 1984 a record-low flood, only about 15 km³ into the Inner Delta; the Niger at Niamey ran dry in June 1985	Up to two-thirds of Tuareg herds lost; families fled Timbuktu for Bassiknou, Mauritania	Lake Faguibine dried and has stayed mostly dry; the UNSO channel project (1986–90) was cut short
August 2003	Torrential rain in Timbuktu, not a river flood	More than 50 homes collapsed in Sankoré and Djingareyber (UNESCO); DFO counted 180	Not recorded
April 2007	The 14.3 km Kabara–Timbuktu canal, funded by Libya, reopened	Not applicable	Now silted and unusable; boats use Koriomé
2008–15	UNEP's Faguibine project, XOF 587 million (US$1 million)	Not applicable	Stalled by insecurity from 2011; the lake was fully dry in 2021
2012–13	Occupation of the north	No Timbuktu toll found	WFP set preconditions for cash in the north that were not met
August–September 2023 (El Niño)	JNIM blockade of Timbuktu from August; the COMANAV boat *Tombouctou* attacked on 7 September	At least 120 dead on the boat (HRW witnesses); food prices up almost tenfold	Supply moved to the river window: Mopti to Koriomé, July–January
April 2024 (El Niño)	Sahel heatwave: Kayes 48.5°C	102 deaths in four days at Gabriel Touré hospital, Bamako, about half over 60 (cause not established)	Not recorded

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
August–November 2024	Record flood: Mopti 5.90 m on 20 August, the highest since 1967; high water at Timbuktu and Diré from mid-November	36,317 affected in Tombouctou region by 9 September; 86–95 dead in Mali	National disaster declared
8 January 2026	An overloaded pinasse sank at night off Diré	33 bodies recovered, 11 survivors	The governor's office urged respect for boarding rules, especially at night
June 2026	Fuel ran out: the power plant stopped on 19 June; no power or piped water in town from 23 June for over a week	No toll reported	One to two hours of power a day per neighbourhood afterwards

The recurrence nobody has noticed: El Niño has never been the killer here on its own. Deaths came when a bad season coincided with relief being cut off: colonial neglect in 1913, the rebellion in 1990, the occupation in 2012, and the blockade in 2023. And every water project worked only while someone dug out the sand each year.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **The danger is heat** (the tracker: 170 forecast deaths) versus **the danger this winter is the supply cut-off** (the Skeptic: only 5 of the 170 fall in December and 13 in January; 61 fall in March). Both hold: the heat deaths fall on the same malnourished children and frail elders as the hunger and infection. The ranking follows the Skeptic, and the heat plan is dated for February.
- **The flood may be high again** (Emergency manager: PRESASS 2026 forecast normal-to-above rain) versus **it may be low** (Skeptic: El Niño tends to weaken Sahel rains). No 2026 gauge reading for Diré was found; the November–December gauges will settle it, which is why action 6 watches both ways.
- **Azithromycin for every child under five** (Public health: −14% mortality in Niger) versus **no benefit on top of SMC in Mali** (verification). Moved to "Do not do this" as a local action.
- **The village is on the supply map** (Food: IMPACT lists "Hondouboumo" among the rural markets Timbuktu households used in the blockade) versus **the match is unconfirmed** (verification: IMPACT places "Hondouboumo et Toya" in the commune of Lafia, whose seat is about 30 km east). Treated as a possible match only.

* Suggested formulation: check with a health worker before sharing.

WHAT NOBODY COVERED

BLIND SPOT

Nothing was found on the village itself: its commune, population, water source, nearest CSCom, or mobile coverage. There is no 2026 Timbuktu price series, no 2026 flood forecast, and no regional death counts by cause. The blockade's status from August to October 2026 is unverified. Nobody in the village was interviewed. Any of these could change the ranking: if the village draws water from the river, action 3's survey matters more; if the blockade has eased, the cash line matters more.

HOW TO READ THIS

METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are published at region scale, for Tombouctou Region. They are not figures for the village. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The location was requested as "Houndou Bongo Keyna" and is taken as Houndou Bongo Koyna, the GeoNames spelling of a populated place at 16.664°N, 2.963°W, about 12 km south-southeast of Timbuktu town, near the Niger River ports of Koriomé and Kabara, in Tombouctou cercle, Tombouctou Region, Mali. Nothing published covers the village itself; its commune and population are unconfirmed. IMPACT (December 2023) lists "Hondouboumo et Toya" (commune of Lafia) among the rural markets Timbuktu households used during the blockade, which is a possible match only. The actions are therefore built at the level of Timbuktu town, the cercle, and the region, and apply to the village by proximity. Mortality figures exist only at region level.
- **Exposure.** Climate Impact Lab El Niño Heat Mortality Tracker data, file `v261026_adm1_output_BASE_rounded_cust.csv` (version of 26 October 2026, pre-peer-review): Tombouctou Region (GADM MLI.9_1), 200 additional heat deaths June–September 2026, 170 forecast October 2026 – March 2027 (October 21, November 46, December 5, January 13, February 21, March 61), 360 in total; Mali, 8,200; the world, 857,400. This update supersedes the September report *451,000 at Risk*. Method: ECMWF SEAS5 seasonal forecasts applied to

* Suggested formulation: check with a health worker before sharing.

temperature–mortality relationships against a 1996–2025 baseline. The figures count heat only; we inferred this from the data, because cold countries show exactly zero in every winter month. They are never added to the Impact Map's long-run projections.

- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources in French and English, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Five verifier agents (health; financial instruments; food and the river; water, river, and heat; the funding envelope) checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist; food actions for published triggers and timing. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted, but the page renders its data in the browser and returned no entries. No precedent was confirmed for any action; that is not evidence that none exists.
- **Money.** Costs are shown in West African CFA francs (XOF) with US dollars in parentheses, at XOF 587.14 = US$1. The rate comes from the fixed peg of XOF 655.957 = €1 and the ECB reference rate of €1 = US$1.1172 (30 October 2026). Figures published in Swiss francs (the Red Cross protocol) are converted approximately. Mali's 10% levy on funding to associations and foundations (decree 2026-0152 of 13 March 2026) may add 10% to NGO-channel lines; whether it applies to UN agencies or the Red Cross is unconfirmed.
- **Reader.** No role was given, so the For you section is written for a resident. Its steps were taken from the verified actions and from what the panel found about households here. Any step not drawn from a ranked action was verified on the same three tests. The rest of the report is written for the region and its partners.

* Suggested formulation: check with a health worker before sharing.

- **Language.** A French version accompanies this one, because French is the language of Mali's administration. Residents speak Songhai (Koyra Chiini), and Tamasheq and Arabic/Hassaniya are also spoken; no version was made in those, and action 7 asks for the key messages to go out on radio in Songhai and Tamasheq instead.
- **Health wording.** Passages where a wording error could hurt someone (malnutrition treatment, meningitis danger signs, cooling and fans, and fasting and the medical exemption) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. This was the second full briefing in one session, so lenses and verifiers were held to about 15 searches each. Two articles on the national response plan and the Food Security Cluster's May 2026 bulletin refused direct reads (403). No 2026 Timbuktu price series, flood forecast, or freight rate was found, and the blockade's status after July 2026 is unverified. The WHO ceftriaxone protocol and the Daugla, Hess, and Meremikwa effect sizes were cited from memory by the lenses and not reopened. The locator map shows the village at country scale.

References · status after verification

Climate Impact Lab, El Niño Heat Mortality Tracker data v261026 (downloaded 30 Oct 2026); tracker · confirmed

River and supply: IMPACT, blockade and Timbuktu markets, Dec 2023; PIARC, Niger navigation; Save the Children, siege of Timbuktu; VOA, blockade prices; FAO GIEWS Mali; WFP Mali Country Brief, May 2026; WFP Mali emergency; UN News, WFP cash preconditions 2012; Purdue, PICS bags; Kabara canal · partially confirmed

Fuel, power, and water: RFI/AllAfrica, 26 Jun 2026; AllAfrica, fuel distribution, 2 Jul 2026; Dakaractu/AFP, a week without water and power; UNICEF, solar at Timbuktu hospital; UNICEF, solar water system; solar borehole appeal price · partially confirmed (solar unit cost: unverified)

Nutrition: Collins 2006, community treatment; Collins, Lancet 2006; UNICEF Mali situation report, Mar 2026; UNICEF, funding shortages; UNICEF Mali, Tombouctou malnutrition; IPC acute malnutrition, Nov 2025 – Oct 2026; GiveWell, CMAM costs; GiveWell–IRC conversation, 2018; ENN, IRC Nara costs; UNICEF RUTF price, 2025; Mali bans French-funded NGOs · partially confirmed (prevalence of 12% and 11.1%: unverified, dropped)

Meningitis and other health: reactive vaccination delay; Men5CV status; WHO meningitis epidemic guidance; Mali-Météo meningitis bulletin, Mar 2025; EDSM-VI 2018; AVENIR trial; Chandramohan 2019; LAKANA, Mali; WHO azithromycin guideline 2020 · partially confirmed (Men5CV in Mali: unverified)

River safety: HRW, Nov 2023; Al Jazeera, 7 Sep 2023; Diré sinking, Jan 2026 · confirmed

Flood: IFRC Mali flood EAP summary; EAP annual report 2025; IFRC DREF MDRML022; Anticipation Hub, Mali; Alwihda, 2024 record; OCHA, Sep 2024;

* Suggested formulation: check with a health worker before sharing.

ReliefWeb, 2024 floods; 2024 flood incidents; Afrik, 2024 toll; UNESCO, 2003 rain damage; DFO 2003 (404 on fetch); PRESASS 2026; AGRHYMET El Niño note 2026 · partially confirmed (riverine framework: unverified beyond listing)

Heat and outreach: Studio Tamani audience study 2025; RSF appeal, Radio Bouctou; Meade 2024, fans; Morris 2021, fans; April 2024 Sahel heat; World Weather Attribution 2024; Ramadan 2027 dates · partially confirmed (Radio Bouctou on air: unverified)

Food and herds: FEWS NET, Sep 2026; FEWS NET, May 2024; APA, PNR 2026; CapMad, PNR 2026; Cash Working Group basket, 2023 · partially confirmed (PNR budget unresolved; transfer value unverified)

Money: APA, humanitarian plan 23% funded; CERF round, Oct 2026; OCHA El Niño briefing, Jun 2026; WFP Sahel layer; ARC Replica payout; Jigisemejiri, closed 2023; UNHAS resumption; ICRC Mali; MSF Mali 2016; 10% levy on associations; ECB reference rate · partially confirmed (ARC 2025–27: unverified)

History: WPF famine table; CSM, Tuareg herds; IWLEARN, 1984 flood; IRD, Niger minimum 1984–85; Lake Faguibine; UNEP Faguibine; Diré (coordinates) · partially confirmed

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

Generated 30 October 2026 for Houndou Bongo Koyna, Tombouctou. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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